Predicting the Yields to Maturity of Senior Secured Bonds

Robert Silaghi, George Kobrossy, Malachi Cassels

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Background

The goal of this analysis is to predict yields for senior secured bonds issued within the past year for companies in the Standard & Poor's 500 index. *Yield to maturity* represents an investor's expected return and is determined by a bond's price, relative to its future payout. By predicting this value, the model helps determine whether a bond is considered cheap or expensive relative to fair value, informing buy or sell decisions. If the predicted *yield to maturity* is greater than the market-implied yield, then the bond is overpriced, and vice versa.

The dataset, collected from FactSet, includes 1,081 rows, where an observational unit is one senior secured bond. Figure 1 and Figure 2 show descriptive statistics. The response variable, *yield to maturity,* is a percentage, which is the unit for interpretation. *Total debt/ebitda* is a unitless leverage ratio, where ebitda is the firm's earnings before interest, taxes, depreciation, and amortization. *Ebitda margin* is a percentage that represents ebitda divided by revenue, measuring operational efficiency. *Market value* is the company size, measured in millions of dollars. *Time to maturity* is a categorical variable that groups the bonds into less than 3 years, 3 to 10 years, and greater than 10 years. *Sales* is the company's revenue and is measured in millions of dollars. Lastly, *rbics economy* classifies issuers into twelve predefined sectors.

Building the Model

The Initial Model

The initial model used to predict the *yield to maturity* of senior secured bonds included *total debt/ebitda*, *ebitda margin*, *market value*, and *time to maturity*. The initial scatterplot matrix and variance inflation factors (VIFs) don't reveal any strong relationships between the predictors and *yield to maturity* (Figure 3). However, after fitting the initial model, the predictor p-values are less than the 0.01 threshold, suggesting that there is very strong evidence that all predictors

have a linear relationship with *yield to maturity* (Figure 4). The R-squared value is 0.3067, so approximately 30.67% of the total variation in *yield to maturity* can be explained using the initial model, showing weak strength (Figure 4). Additionally, the residual standard error is 0.95, so the size of the typical error is 0.95%, given *yield to maturity* is expressed as a percentage (Figure 4).

In the residual plot, the residuals appear mostly randomly scattered with no clear curve, but those around the middle fitted values have a larger spread (Figure 4). Therefore, linearity is met, but the constant variance assumption is not. In the QQ-plot, points deviate from the identity line for extreme theoretical quantiles, violating the normality assumption (Figure 4). However, the large sample size (n = 1081) allows the sampling distribution of the estimated regression coefficients to remain approximately normal, so inference is still appropriate. Lastly, while the independence assumption is likely reasonable as the data is primarily cross-sectional, some companies have multiple bond offerings. This could be problematic as unexplained firm characteristics might cause dependence. However, most firms only have one offering in the dataset, meaning that independence can be reasonably assumed.

Log Transformations

Log transformations were considered for variables with a right skew. *Yield to maturity* and *ebitda margin* did not benefit from a log transformation (Figure 5, Figure 6). *Market value* has a right skew and benefited from a log transformation (Figure 7). This transformation revealed a linear relationship between log(*market value*) and *yield to maturity* (Figure 11). *Total debt/ebtida* was evaluated for log-transformation; however, this complicates the interpretations and R-squared did not improve (Figure 9, Figure 10).

Additional Predictors

Sales was considered as an additional predictor, but no significant relationship was found ($p = 0.55$, Figure 13). Additionally, introducing *sales* results in a VIF of 2.12, suggesting some multicollinearity (Figure 13). Combined with weak predictive power, *sales* was not considered further. *Rbics economy* was also explored. Despite 11 coefficients being added, the adjusted R-squared increased from 0.3250 to 0.3495 (Figure 10, Figure 14). However, only the Consumer Services sector's p-value showed strong evidence of a difference in the *yield to maturity* compared to the reference sector, Business Services (Figure 14). While the other sectors showed some or no evidence in favor of the alternative, the residual plot including *rbics economy* exhibited more homoscedasticity for middle-of-the-range fitted values (Figure 14). A nested F-test resulted in a p-value < 0.01, suggesting that at least one *rbics economy* coefficient differs from 0 (Figure 15). Lastly, Mallow's Cp was considered, with the full model consisting of 23 predictors (Figure 16). The model with *rbics economy* had a value of 79.56, while the one without was 112.02 (Figure 16). Therefore, *rbics economy* was included.

Interaction

Lastly, an interaction between *time to maturity* and *debt/ebitda* seemed plausible (Figure 17). The QQ-plot was roughly unchanged again, while the residuals were more homoscedastic (Figure 18). The nested F-test supports the addition of the interaction ($p = 0.02$, Figure 19). Furthermore, when including the interaction, Mallow's Cp decreased from 79.56 to 74.83 (Figure 20). Thus, it was included in the final model.

Model Results

The Final Model

The final model predicts *yield to maturity* using *total debt/ebitda*, *ebitda margin*, log*(market value)*, *rbics economy*, *time to maturity*, and an interaction between *time to maturity* and *total debt/ebitda* (Figure 18). The residuals are mostly randomly scattered with no clear curve, and residual variability was approximately constant except for slightly larger spreads for some middle fitted values (Figure 18, Figure 21, Figure 22). Therefore, the linearity and constant variance assumptions are reasonable. For the QQ-plot, points still deviate from the identity line for more extreme theoretical quantiles, violating the normality assumption (Figure 18). However, the large sample size still allows for statistical inference. The independence assumption is also reasonable, for the same reasons as the initial model.

Model Strength

The final model R-squared observed is 0.3641, which means that approximately 36.41% of the total variation of yield to maturity can be explained using the final linear model (Figure 18). Additionally, the final residual standard error is 0.9182, meaning that the size of a typical error when predicting *yield to maturity* is 0.92%. In the original histogram of the dependent variable, the standard deviation was 1.14% (Figure 1). Using this information, the model can be subjectively interpreted as moderately strong.

Multicollinearity and Overfitting

In the final model, there is no evidence of multicollinearity. The largest correlation is -0.34 between log(*market value*) and *total debt/ebitda*, which is not problematic (Figure 12). Additionally, the largest VIF is 1.18 for the log(*market value*), which is sufficiently low. The scatterplot matrix above also does not illustrate any visual trends of multicollinearity, as points

seem to be randomly scattered between predictors (Figure 12). Furthermore, each category has at least 10 observations, so overfitting is not a concern (Figure 1).

Predictor Inference

Time to maturity is practically significant; when compared to long-term maturities, *yield to maturity* decreases by 0.81% for medium-term and 1.47% for short-term maturities for companies with *total debt/ebitda* = 0, holding other predictors constant (Figure 18). This makes sense as longer time frames are more uncertain, leading to higher expected compensation. Log(*market value*) also shows practical significance, since as *market value* increases by 100%, ranging from $6.45 billion to $3.75 trillion, the *yield to maturity* decreases by 0.47%, holding other predictors constant (Figure 1, Figure 18). Since larger companies are stable, this minimizes risk and, therefore, decreases yield. The coefficients for the *rbics economy* sectors vary in practical significance. For example, when holding other predictors constant, *yield to maturity* for consumer services is 0.49% higher, on average, than business services (Figure 18). *Total debt/ebitda* is also practically significant; as it increases by one unit, ranging from 0.12 to 19.16, *yield to maturity* increases by 0.11%, holding other predictors constant (Figure 1, Figure 18). This occurs because debt leads to higher bankruptcy risks. Lastly, the negative interaction coefficients mean that as *total debt/ebitda* increases by one unit, *yield to maturity* increases 0.10% more for long-term durations than medium and short-term ones, holding other predictors constant. This is practically significant as the coefficient for *total debt/ebitda* for long-term durations (the reference category) is 0.11 (Figure 18). This shows compounding uncertainty over time.

Conclusion

The goal of this analysis was to explore whether the yields for senior secured bonds for companies in the Standard & Poor's 500 index can be predicted using an ordinary least squares model. It was moderately successful at accomplishing this goal, as the average error was reduced by 19.47% and the model explained 36.41% of the variation in *yield to maturity*. Logical and practical relationships appear that higher *debt/ebitda*, longer *times to maturity*, and smaller *market values* consistently predict higher *yields to maturity*. A concern with the independence assumption was mitigated by diversifying the dataset to include many different bond-issuing companies with mostly just one observation.